

NEW BOOKS.

Traité Élémentaire de Mécanique Chimique fondée sur la Thermodynamique. P. DUHEM. Two volumes. Paris, 1897.

This excellent book gives a systematic treatment of the applications of thermodynamics to chemistry at a time when current memoirs are already widely departed from the elements of the subject and the need of such systematic treatment is seriously felt by students of physical chemistry. It is particularly gratifying that one so preëminent among specialists in "chemical mechanics" should undertake the work, giving us a treatise which may "fairly be regarded as the best extant exposition of the subject which it treats."

The first volume is devoted to the general theory of thermodynamics and to the theory of what Duhem calls false equilibria. The second volume is devoted to vaporization and analogous phenomena and to dissociation.

The foundations of mathematical theory should always be those things which are sensible and measureable, never the "supposed smoothness of molecules or other rubbish of this fancy sort." Duhem's treatise meets the most exacting requirements in this respect.

Applied algebra, however, by no means embraces the whole range of physical theory; indeed concrete theories, such as the molecular theory, are much more important to the beginner than the purely algebraic theory. The beginner must master ideas, images of things, sufficiently abstracted to meet in the mind. Mathematical theory and concrete theory are of course not to be compared, being supplementary to each other. The latter has mainly to do with discoveries and applications, while the former follows with refinements and elaborations.

The writer is of the opinion that for teaching purposes, even for advanced students, an abstracted algebraic theory is less effective than analysis¹ in which ideas play a dominant part. Maxwell's classical treatise on electricity and magnetism is perhaps the best example we have of the latter style. There is no doubt a certain mathematical incompleteness in such treatise, but it seems that the various mathematical treatises which have been written with a view to the elucidation of Maxwell should follow the study of Maxwell to clear up the vexatious algebraic

¹ The writer has questioned a number of the leading American teachers of physics and engineering and is surprised to find that the prevalent notion of "analytical mechanics" seems to be that it is the formal application to mechanics of algebra including calculus!

questions of limits, continuity, and invariance, for the writer's experience is that almost any student having an elementary knowledge of physics and mathematics can, with some help from a teacher, fully comprehend Maxwell's treatise. Surely it will not always be necessary for students to reestablish the fundamental notions in physics by algebraic analysis when a few generations of experience transcends all algebraic analysis in its substantial verification of the notions which we apply to the physical world around us; and it seems that concrete theory—that help for the mind corresponding to tools for the hand—must continue to grow in its range and importance. At present, however, the manifest inadequacy of our notions of electricity and magnetism is such that the purely algebraic aspects of our theories must be fully examined by a student before he has mastered the present state of the science. The writer cannot refrain in this connection from a word in praise of Webster's recent treatise on the mathematical theory of electricity and magnetism.

Thermodynamics, however, does not involve such a wide variety of geometrical and physical notions as does the subject of electricity and magnetism, and the theory of thermodynamics is perhaps to remain purely algebraic. However this may be, the writer does not share the author's regrets in being "*décidé à renoncer à une exposition absolument rigoureuse des principes de la Thermodynamique*."

It seems to the writer that the application¹ of the summation $\sum \frac{dQ}{T}$ to an irreversible or sweeping process is erroneous for the reason that, during a sweep, a system has no temperature since temperature is a mark only of a state of equilibrium. The error of this application has indeed been pointed out by Gibbs. However, the summation $\sum \frac{dQ}{T}$ has a definite significance in connection with a sweeping process as follows: Consider a sweep A which brings a system from a state of equilibrium 1 to a state 2, and a reversible process B which brings the system from 2 to 1. The above summation may then be applied to this process B , which is in a sense the reverse of A , giving $\sum_B \frac{dQ}{T} < 0$. Indeed the summation $\sum \frac{dQ}{T}$ may find application during the process A in case an adjoining system undergoes a reversible process under the influence of a sweep A , namely:

$$\sum \frac{dQ}{T} - S_1 + S_2 = P^2 \quad (1)$$

in which S_1 and S_2 are the values of the entropy³ of the given system in

¹ Duhem, Vol. I., p. 81.

² Equation (11). Duhem, Vol. I., p. 82.

³ That is $S_2 - S_1 = \sum_B \frac{dQ}{T}$ in which B is the process before mentioned.

its initial and final states 1 and 2 respectively, P is a positive quantity, and $\sum \frac{dQ}{T}$, applied to the external system which undergoes a reversible process under the influence of the sweep A , is the increase of entropy of that system. The results of Duhem's analysis are not affected by his ostensible application of the summation $\sum \frac{dQ}{T}$ to a sweeping process; the only effect of the error, if such it really is, is to confuse the reader and to give apparent support to some statements made by the author regarding thermodynamic potential.

Duhem's claim¹ that the use of the thermodynamic potential is less ambiguous than the use of the entropy (method of Horstmann) in the study of a given system in that it takes more explicit account of the action of external systems, seems to the writer to be unwarranted. This idea grows out of the application of the summation $\sum \frac{dQ}{T}$ to a system during a sweeping process.

A sweeping process is essentially independent of external action upon the system which undergoes the sweep and a system may undergo any given sweep without exchanging heat or work with external systems. This is evident when we consider that a sweep is the passage of a system from an unstable state to a state of stable equilibrium. An apparent exception is afforded by any system subject to external action which changes at a finite rate. In this case the system falls behind as it were. A real exception is the sweeping process which iron undergoes as it is magnetized, but it must be remembered that the magnetic state of iron cannot be specified by a finite number of generalized coördinates and is therefore not amenable to the principles of thermodynamics as we at present know them. Perhaps the process of crystallization is a real exception also. If so, then the state of a system consisting of crystals and a mother liquor cannot be completely specified by a finite number of generalized coördinates. The question in regard to crystallization is simply this: Can a crystal be made to grow at an infinitely slow rate positive or negative and if so are the successive aspects of the system during the slow growth of a crystal precisely similar to the successive aspect of the system during the slow dissolution of the crystal? The phenomenon of etching of crystals seems to indicate that this is not the case, but, of course, the condition necessary to the production of etch figures is that the dissolution of the crystal be as rapid as possible. It would be interesting to study the forms of crystals under the condition of slow positive and negative growth. Indeed, the writer has undertaken to carry out observations along this line.

¹ Vol. I., p. 88.

Systematized physical science treats of phenomena associated with states of equilibrium or with material conditions which may be characterized by a finite number of definite (corresponding) elements, generalized coördinates,¹ generalized forces, rates, etc., such as volumes, pressures, reaction velocities, and the like. When a given set of physical conditions is reduced to its elements the science of the thing is accomplished, tentatively if the elements are not fundamental. Every bit of systematic knowledge in the whole range of the physical sciences exemplifies this fact.

It seems that a complete method for the study of sweeping processes, in the widest sense of that term, cannot be built up from our present methods of thought at the bottom of which is the idea of correspondence, or of cause and effect, if one so chooses to call it; for in the general aspect of the sweeping process this principle of correspondence fails. The decrease in thermodynamic motivity (Lord Kelvin) during a sweep constitutes an activity the results of which may be made to diverge widely by some infinitesimal circumstance which occurs during the initial stages of the sweep. This is shown by any system which sweeps from a state of unstable equilibrium into a state of stable equilibrium. Many sweeps do not indeed take their rise in states of unstable equilibrium, for example the sweeping process which a substance undergoes while the external actions on the substance change at a finite rate; such sweeps are reducible to elements among which the rates of change of the external actions appear. The most general type of sweep is that which brings a system from a state of unstable equilibrium to a state of stable equilibrium. The changes of the atmosphere which constitute weather phenomena are the most familiar examples of the general type of sweep. A state of unstable equilibrium is produced by the heating of the lower strata of the atmosphere. An infinitesimal action, as the waving of a fan, may precipitate a sweep, and slight variations of the circumstances, time and place of the beginning of the sweep may produce modifications in the progress of the sweep which are absolutely incommensurate to these causes, or in other words, an infinitesimal cause may produce a finite effect.

Long range detailed weather prediction is therefore impossible, and the only detailed prediction which is possible is the inference of the ultimate trend and character of a storm from observations of its early stages; and the accuracy of this prediction is subject to the condition that the flight of a grasshopper in Montana may turn a storm aside from Philadelphia to New York!

¹ Energy—the kind we think about—is nothing more than a generalized notion not to be properly thought of as existing outside of one's head except for the sake of concreteness.

A sweeping process—and no others are realizable—is then more or less a thing of chance, if one may use this word to characterize the failure of the principle of correspondence. This same conclusion is reached by the molecular theory in the study of gases. Indeed, it seems that the atomistic method is the only method capable of being extended to the study of sweeping processes, for where the different parts of a substance are in different states a study of the substance requires explicit consideration of these parts, but of course this is in general impossible. In those cases in which the behavior of a substance may be represented by a finite number of elements the substance must be homogeneous, or composed of a number of homogeneous phases; and every one of these phases, if not in thermal equilibrium, must deviate therefrom in a manner which is precisely the same¹ throughout its mass. Examples of such simple sweeping processes are to be found in the various cases in which reactive velocities have been studied.

Whether a method can be established for studying a substance by infinitesimal parts by the use of infinite series, which by more or less extended summation give approximate results, cannot be predicted, although it seems that simple sweeps might be followed in this way. For example, it would not be difficult to formulate the behavior of a gas expanding in a cylinder against a piston moving at a given velocity.

It can be shown from the principles of thermodynamics, as we know them, that the condition of thermal equilibrium, namely, that the variation of entropy for any virtual change in the system is zero or negative, must always have a unique solution so that only one state of equilibrium is possible for given external actions on a system. Now the most striking case in which the fundamental assumption² of thermodynamics fails is the case of magnetized iron, and Ewing has shown that the facts in this case may be represented by assuming that the process of magnetization is a process of which the successive aspects are states of equilibrium, infinite in number, separated by intervening sweeps; and the writer is inclined to think that the extension of the principles of thermodynamics which will finally include magnetization, and what Duhem calls *false equilibria*, will be an extension which admits the possibility of more than one solution of the condition of equilibrium. Indeed, the writer is inclined to look upon the conceptions of viscosity and friction which Duhem makes use of in his treatment of false equilibria as being as fanciful as, for example, the assumed smoothness of molecules—except, and it is a very essential

¹In general, the deviation, at a point, from homogeneity and from thermal equilibrium must be known functions of the point. It is only those cases in which this function is of the simplest form that have been reduced to their elements.

²That the state of a system in equilibrium may be specified by a finite number of generalized coördinates.

exception, that the conceptions of viscosity and friction apply to things large enough to be felt and seen. Duhem's treatment of false equilibria may, therefore, be a false treatment of true equilibria. This, however, no man knows, and the treatment which Duhem gives is extremely valuable, being, as it is, the first systematic treatment of these phenomena.

This treatise of Duhem's is about as satisfactory as it is possible for such a treatise to be in the present state of our knowledge of thermodynamics.

W. S. FRANKLIN.

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Elementary Text-Book on Physics. By WM. A. ANTHONY and CYRUS F. BRACKETT. Revised by WILLIAM FRANCIS MAGIE. 512 pp. John Wiley & Sons, New York, 1897.

The supreme test of merit in a book whose object is to set forth the principles of a science is to be found in its *teaching value*. If this is a truism—and it reads like one—it is not always apprehended. We have so many class-room texts and of such varying worth, so many cases of much meritorious matter hopelessly involved in the subtleties of its own expression, and (alas!) so many instances in which a professed desire to instruct has been over-ridden by unconscious pedantry, that a moment's consideration of the subject, at this time, may not be inappropos. If my remarks are construed as championing the cause of the student, I shall not complain. It is, perhaps, time that this worthy were spoken for.

As a piece of literature, the "text-book" must ever occupy a limited sphere. Essentially didactic and in no degree creative, it may possess only the sterner graces of composition. There must be careful diction and concise phraseology. There must be logical sequence and, as to substance, always, of course, strict adherence to scientific truth. But in order that a book may be in the highest degree *teachable*—if I may use the term objectively—it must possess certain more subtle and indefinable qualities as a part of its birthright. With the limitations of style referred to, the writing of a scientific text-book would seem a prosaic enough task—one furnishing few opportunities for the impress of the author's personality. And yet I venture to assert that two treatises on the same branch of science, the individual products of two men of fairly equal intellectual attainments, will differ markedly in what I have called *teaching value*. After all, the work is likely to be the child of the man. "Your born teacher is as rare as a poet, and as likely to die young," writes a member of the pedagogical calling. Quite as rare, and perhaps as ephemeral, is the class-book in which is combined unquestioned scientific merit and the fullest power of yielding up the matter between its